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1 Introduction

The application to be considered relies on a collection of movies. A sample of the data sets is illustrated by figure 1. The attributes of a movie are two identifiers (id and idimdb), the release data, the runtime, the budget and revenues, a link to a poster, an overview and a rating.

id	title	release	runtime	budget	revenue	poster	overview	rating	last_update
tt0029583	Snow White and the Seven Dwarfs	1937-12-21	83	1488423	184925486	https://image.tmbd.org/t/p/original/1P9eGGITvE7vKA...	A beautiful girl, Snow White, takes refuge in the ...	7.6	2022-02-27
tt0032910	Pinocchio	1940-02-23	88	2600000	84300000	https://image.tmbd.org/t/p/original/ynKEiE296f9sK9...	Lonely toymaker Geppetto has his wishes answered w...	7.4	2022-02-27
tt0033563	Dumbo	1941-10-31	64	812000	1600000	https://image.tmbd.org/t/p/original/hKDDllisMtsU9J...	Dumbo is a baby elephant born with over-sized ears...	7.2	2022-02-27
tt0034492	Bambi	1942-08-14	70	858000	267447150	https://image.tmbd.org/t/p/original/wV9e2y4myJ4KMF...	Bambi's tale unfolds from season to season as the ...	7.3	2022-02-27
tt0042332	Cinderella	1950-02-22	74	2900000	263591415	https://image.tmbd.org/t/p/original/4nssBCQubADCTB...	Cinderella has faith her dreams of a better life w...	7.3	2022-02-27
tt0046183	Peter Pan	1953-02-05	77	4000000	87404651	https://image.tmbd.org/t/p/original/1JJOs1yrhKfZc...	Leaving the safety of their nursery behind, Wendy,...	7.3	2022-07-13
tt0048280	Lady and the Tramp	1955-06-22	76	4000000	36359037	https://image.tmbd.org/t/p/original/wXKeGea4htM7Yz...	Lady, a golden cocker spaniel, meets up with a mon...	7.3	2022-10-18
tt0053285	Sleeping Beauty	1959-02-17	75	6000000	51600000	https://image.tmbd.org/t/p/original/2mTyUukLwDle...	A beautiful princess born in a faraway kingdom is ...	7.2	2022-02-27
tt0055254	One Hundred and One Dalmatians	1961-01-25	79	4000000	85000000	https://image.tmbd.org/t/p/original/9kDiS1sVb5Ld...	When a litter of dalmatian puppies are abducted by ...	7.3	2022-02-27
tt0057546	The Sword in the Stone	1963-12-25	79	3000000	22182353	https://image.tmbd.org/t/p/original/7lyeeuhGAJSNXY...	Wart is a young boy who aspires to be a knight's s...	7.2	2022-02-27
tt0058331	Mary Poppins	1964-08-27	139	6000000	103082380	https://image.tmbd.org/t/p/original/yO8UyO86uNBq12...	A magical nanny employs music and adventure to hel...	7.8	2022-02-27

Figura 1: Sample

2 Data representation and hashing

American Standard Code for Information Interchange (ASCII) is a computer standard for character encoding. Figure 2 illustrates a representation of the ASCII table.

						0 0 0	0 0 1	0 1 0	0 1 1	1 0 0	1 0 1	1 1 0	1 1 1
						0	1	2	3	4	5	6	7
↓	↓	↓	↓	↓	0	NUL	DLE	SP	0	@	P	\	p
0	0	0	1	1	1	SOH	DC1	!	1	A	Q	a	q
0	0	1	0	2	2	STX	DC2	"	2	B	R	b	r
0	0	1	1	3	3	ETX	DC3	#	3	C	S	c	s
0	1	0	0	4	4	EOT	DC4	\$	4	D	T	d	t
0	1	0	1	5	5	ENQ	NAK	%	5	E	U	e	u
0	1	1	0	6	6	ACK	SYN	&	6	F	V	f	v
0	1	1	1	7	7	BEL	ETB	'	7	G	W	g	w
1	0	0	0	8	8	BS	CAN	(	8	H	X	h	x
1	0	0	1	9	9	HT	EM	)	9	I	Y	i	y
1	0	1	0	10	10	LF	SUB	*	:	J	Z	j	z
1	0	1	1	11	11	VT	ESC	+	;	K	[	k	{
1	1	0	0	12	12	FF	FS	,	<	L	\	l	
1	1	0	1	13	13	CR	GS	—	=	M	]	m	}
1	1	1	0	14	14	SO	RS	.	>	N	^	n	~
1	1	1	1	15	15	SI	US	/	?	O	_	o	DEL

Figura 2: USASCHII

2.1 Exercise 1

We assume a hashing function consisting in the sum of the ASCII code of the characters. Process the hash values for the following movies: Rio, Home, Soul, Cars, Luca, Wish, Sing, Bolt, Dune, Vivo, Coco, Dumbo, Brave, Se7en, Mulan and Shrek.

3 Distributed hash table

A Distributed Hash Table (DHT) is a distributed system relying on a lookup mechanism similar to the one in hash tables. Key-value pairs are stored in a DHT, and any participating node can efficiently retrieve the value associated with a given key.

3.1 Exercise 2

1. Let a hash function $\text{mod } n$, where n is the number of servers, to be used in a DHT. If n equals to 3 with servers $A \pmod{3} = 0$, $B \pmod{3} = 1$ and $C \pmod{3} = 2$, give the distribution of the previous movies.
2. We assume server C leaves the network. Give the new distribution of the elements in the DHT (with A associated to $\text{mod } 2 = 0$ and B to 1). Give the number of modifications.
3. We assume server D integrates the network (with A, B and C). Give the new distribution of the elements in the DHT (with A associated to $\text{mod } 4 = 0$, B to 1, C to 2 and D to 3). Give the number of modifications.

4 (Simple) Consistent hashing

Consistent hashing is a hashing technique limiting the number of elements to be remapped when the structure is resized.

4.1 Exercise 4

1. We assume Server A to be associated to the value (in degrees) 120, B to 240 and C to 360. Give the distribution of the movies on the cluster.
2. We assume server C leaves the network. Give the new distribution of the elements. Give the number of modifications.
3. We assume a (very) simple distribution of some virtual nodes. Give the distribution of the movies.

Server A:	Server B:	Server C:
-----	-----	-----
A0: 0	B0: 12	C0: 24
A1: 36	B1: 48	C1: 60
A2: 72	B2: 84	C2: 96
A3: 108	B3: 120	C3: 132
A4: 144	B4: 156	C4: 168
A5: 180	B5: 192	C5: 204
A6: 216	B6: 228	C6: 240
A7: 252	B7: 264	C7: 276
A8: 288	B8: 300	C8: 312
A9: 324	B9: 336	C9: 348

4. We assume server C leaves the network. Give the new distribution of the elements.
5. We assume server D integrates the network (with A, B and C). Give the distribution of the movies.

ServerD:

D6: 23
D0: 47
D7: 89
D1: 103
D8: 145
D2: 169

D9: 211
D3: 221
D4: 286
D5: 349

6. We assume a strategy for randomized positions. This randomized and interleaved distribution ensures that if one server leaves, the remaining servers will more evenly share the load, avoiding a disproportionate shift to a single server. Give the distribution of the movies.

A0: 0
B0: 15
C0 : 29
A1: 43
B1: 58
C1 : 72
A2: 87
B2: 102
C2 : 115
A3: 130
B3: 145
C3 : 158
A4: 173
B4: 188
C4 : 202
A5: 217
B5: 232
C5 : 245
A6: 260
B6: 275
C6 : 288
A7: 303
B7: 318
C7 : 332
A8: 347
B8: 362
C8 : 375
A9: 390
C9 : 418
B9: 405

7. We assume server C leaves the network. Give the new distribution of the elements.
8. We assume server D integrates the network (with A, B and C). Give the distribution of the movies.

D0 : 47 degra
D1 : 103 degra
D2 : 169 degra
D3 : 221 degra
D4 : 286 degra
D5 : 349 degra
D6 : 23 degra
D7 : 89 degra
D8 : 145 degra
D9 : 211 degra

5 References

- **Lecture:** https://people.irisa.fr/Laurent.Dorazio/data/teachings/big_data/big_data_cassandra.pdf
- **Cassandra:** https://cassandra.apache.org/_/index.html